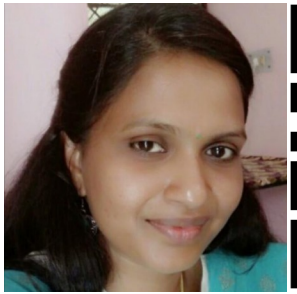
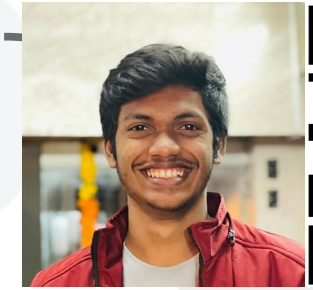
GEOSMART INDIA
2024

Bridging the Gap: Integrating Mesh Networks with LEO Satellites for Improved Remote Area Connectivity



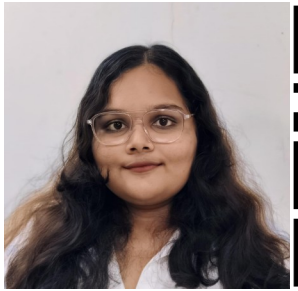
Prof. D Santhadevi



Tatikonda Karthikeya

Dakshin Sanjay

Krishna Reddy



Gayathri M.

Dilnasha M.

Connecting Towers to LEO Satellites: Transforming Remote Connectivity

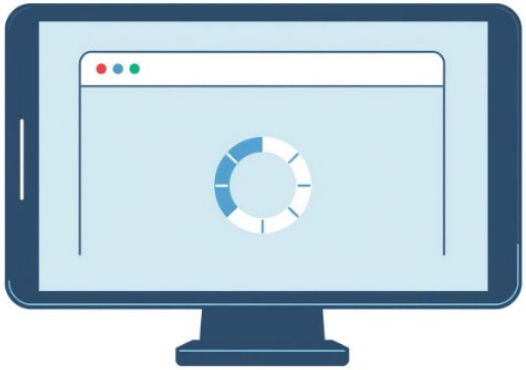
- Connectivity Challenges
- LEO Satellite Solutions
- Transformative Impact

Challenges in Remote Connectivity

- High Deployment Costs
- Slow Rollout Times
- Infrastructure Gaps
- Objective
- Current Solutions

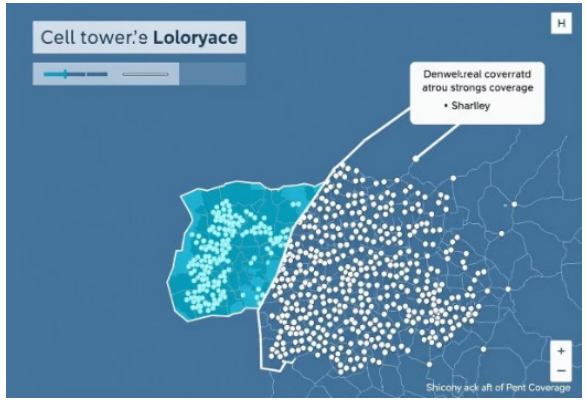
Issues with Traditional Cellular Networks

High Latency

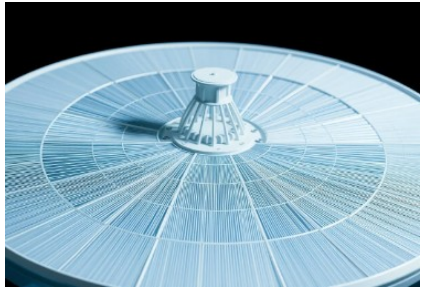


Costly Infrastructure

Limited Coverage



Tower Upgradation with Individual Costs



Antennas

Transceivers

Cooling Systems

Power Systems

Satellite - Cell Tower Integration System

Key Benefits :

- Expanded Coverage
- Improved Reliability
- Future-Ready

Addressing the Limitations

Power Demand

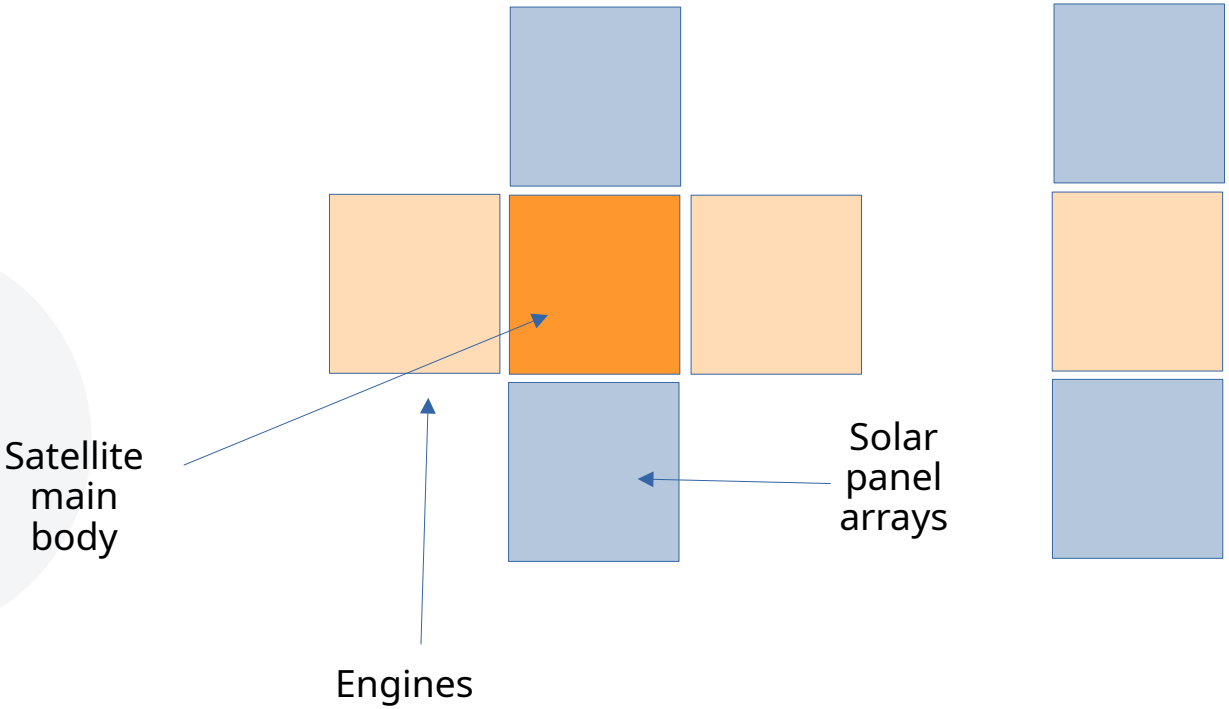
Maintenance

High Initial Costs

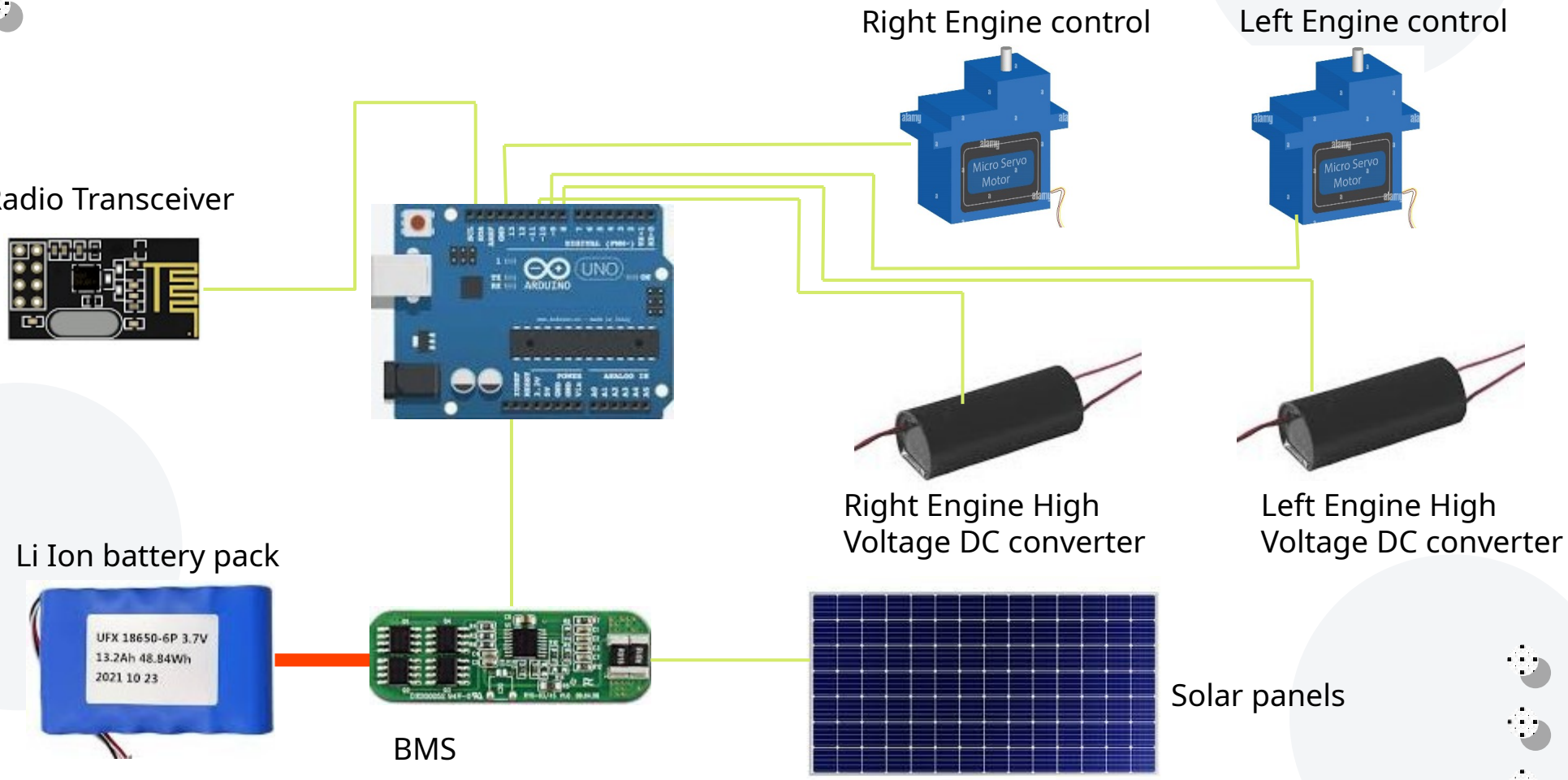
Technology Demonstrator Satellite:

Front view:

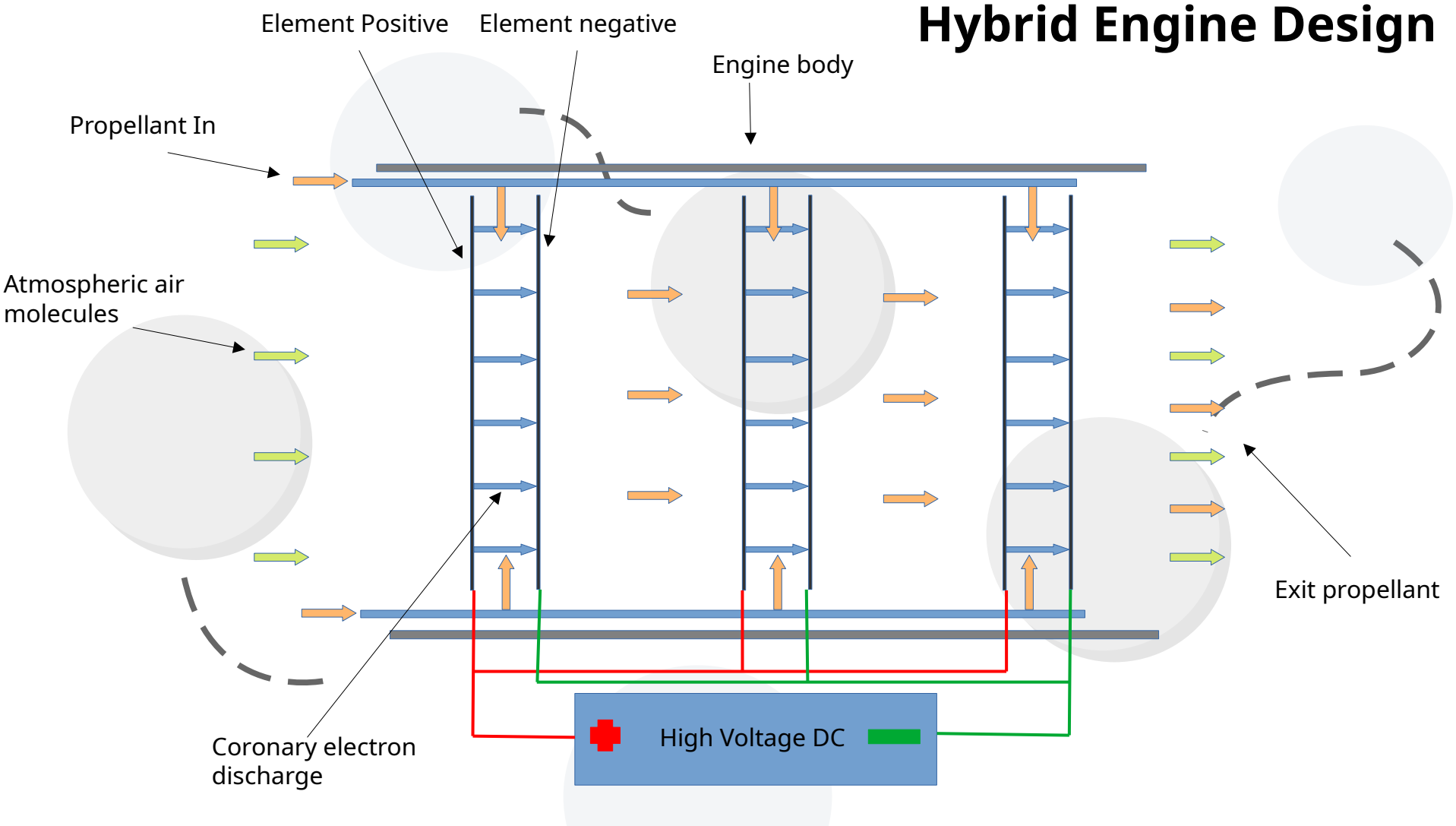
Side/Top view:

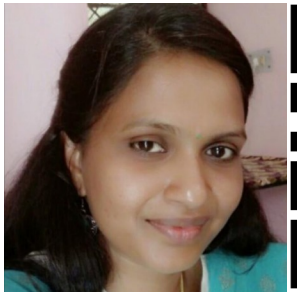


Satellite Electronics:

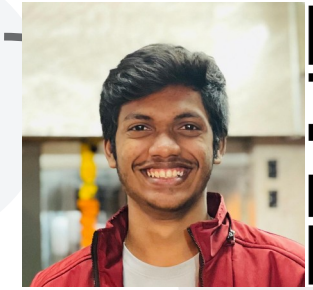


Hybrid Engine Design





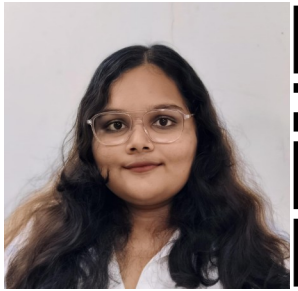
Prof. D Santhadevi



Tatikonda Karthikeya

Dakshin Sanjay

Krishna Reddy



Gayathri M.

Dilnasha M.